

The Octonionic Shadow of GF(7):

Color, Chirality, and Three Generations
from a Quadratic-Residue Difference Set

Nova Spivack

Independent Researcher

July 2026

Preprint.

Archival version (Zenodo): <https://doi.org/10.5281/zenodo.21200048>

UGP series hub (Zenodo): <https://doi.org/10.5281/zenodo.20168144>

© Nova Spivack 2026. Licensed under [Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International \(CC BY-NC-ND 4.0\)](#).

Classification: Theoretical Physics; Algebraic Geometry; Group Theory; Division Algebras; Standard Model.

Abstract

The UGP/GTE arithmetic programme derives Standard Model structure from GF(7) and the group $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$, with major claims machine-certified in Lean 4. The division-algebra programme (Dixon; Furey; Baez–Huerta) independently derives one-generation Standard Model structure from the octonions. This paper proves the two programmes are facets of a single finite structure: the quadratic-residue set $\text{QR}(7) = \{1, 2, 4\} \subset \mathbb{F}_7^*$ is simultaneously the arithmetic anchor of GTE and the Fano-plane orientation that defines the octonion algebra.

Starting from $D = \{1, 2, 4\}$ as the sole input, we establish a six-link machine-verified derivation chain: (i) the oriented translates of D are the octonion multiplication table, uniquely up to relabeling, with the sign encoding $\text{sign}(e_x e_y) = [y - x \in D]$ read directly from the difference set; (ii) $N_c = 3$ from the pencil count of the $(7, 3, 1)$ design via $F_{21} \hookrightarrow G_2$ and $\text{Stab}_{G_2}(\text{apex}) = \text{SU}(3)$; (iii) the exceptional isomorphism $\text{PSL}(2, 7) \cong \text{GL}(3, 2)$, order 168, certified by Todd–Coxeter coset enumeration; (iv) the derivation algebra $\text{der}(\mathbb{O}) = \mathfrak{g}_2$ (dimension 14 exact) and $\text{Stab}_{G_2}(\text{apex}) = \mathfrak{su}(3)$ (dimension 8 exact), with bracket closure and Killing-form certificate computed from first principles ($\mathbf{A}_{\text{Lean}}$, 39 Lean theorems); (v) UGP’s flavor package (μ_3, U, V_4, A_4) is equivariantly isomorphic to the Spin(8) triality package $(\rho, \sigma, Z(\text{Spin}(8)), Z \rtimes \rho)$, pinning $\text{gen}_1 \leftrightarrow V$ via Eisenstein norm uniqueness $b_{\text{gen}_1} = 73 = \mathcal{N}(9 + \omega)$ ($\mathbf{A}_{\text{Lean}}$); and (vi) the charged-lepton Koide mass ladder is a function on the triality orbit with torsor-invariant angle $|\delta \bmod 2\pi/3| = 0.222229631 = 2/9 + 7.4 \times 10^{-6}$ (7 ppm from PDG).

Two structurally independent derivation paths — finite arithmetic (GTE) and continuous division algebra — converge on identical outputs (N_c , charge spectrum, $N_{\text{gen}} = 3$); this convergence constitutes non-trivial structural evidence that these quantities are genuine consequences of the GF(7) arithmetic rather than numerical accidents. The neutrino sector is completed by the GTE seesaw: the right-handed seed ordering $b_{R,1} = 5 < b_{R,2} = 11 < b_{R,3} = 19$ pins the mass eigenstates ν_k via $m_{\nu,k} \propto b_{R,k}^{29/9}$ and predicts normal ordering, with Δm_{21}^2 and Δm_{31}^2 within 1% of NuFIT 6.0 IC24 NH. All major claims are certified at level $\mathbf{A}_{\text{Lean}}$ (zero **sorry** in Lean 4) or $\mathbf{A}$ (computation-verified by assertion-checked Python scripts). §10 lists what is not established.

Contents

1	Introduction	3
2	GF(7) and the Fano Plane	5
3	The Octonion Construction	5
4	The Group Layer	6
5	Color Sector and the UGP Interface	7
6	The Electroweak Shadow	8
7	Three Generations from Triality	9
8	Relation to the Division-Algebra Programme	14
8.1	Programme hierarchy	14
8.2	Comparison with Furey–Hughes (2025)	15
8.3	Original contributions	15
8.4	Interpretive significance	16
9	Formalization Programme	17
10	Open Problems and Negative Results	18
	Appendices	19
A	Claim-to-Evidence Matrix	20
B	Lean Module Inventory	22

Claim-strength taxonomy

Every result in this paper is tagged with its confidence level:

- **A_{Lean}** — machine-certified in Lean 4 with zero `sorry`. All cited Lean theorems reside in the `ugp-lean` library.
- **A/D** — full analytic derivation; computationally confirmed.
- **A** — computationally verified (assertion-checked Python script); analytic derivation or Lean certificate remains open.
- **B** — plausible and partially supported.
- **D** — exploratory or conjectural.

Ordering: **A_{Lean}** > **A/D** > **A** > **B** > **D**. Lean theorem names appear in `monospace`; all Lean modules cited here are zero-sorry and reside in `ugp-lean`. Level framing: Theorems A–H are Level 0–1 algebraic certificate claims (GF(7) arithmetic and F_{21}/G_2 group theory). The C3'' Lagrangian frontier is Level 3; open items in §10 are labeled accordingly.

What this paper claims

Algebraic certificates (**A_{Lean}**/**A**):

- [**Octonion from QR(7)**] $D = \{1, 2, 4\}$ orients the Fano plane, giving the octonion multiplication table uniquely; 480 of 3840 oriented tables are composition algebras (**A**, `octonion_from_qr7.py`).
- [**Group certificate**] $|\mathrm{PSL}(2, 7)| = 168$, certified by 168-coset Todd–Coxeter table (**A_{Lean}**, `hurwitz_presentation_order_168`).
- [$F_{21} \hookrightarrow G_2$] UGP’s gauge skeleton embeds in $G_2 = \mathrm{Aut}(\mathbb{O})$ as octonion automorphisms; $N_c = 3$ from pencil count (**A_{Lean}**, `Nc_factorization`, `weights_card_from_pencil`).
- [G_2 and **SU(3) derived**] $\mathrm{der}(\mathbb{O}) = \mathfrak{g}_2$ (dim 14) and $\mathrm{Stab}_{G_2} = \mathfrak{su}(3)$ (dim 8) from exact linear algebra (**A_{Lean}**, `derivation_algebra_is_g2_certificate`, `apex_stabilizer_is_su3_certificate`).
- [**Triality = flavor**] UGP’s (μ_3, U, V_4, A_4) is equivariantly isomorphic to $\mathrm{Spin}(8)$ triality $(\rho, \sigma, Z, Z \rtimes \rho)$; $\mathrm{gen}_1 \leftrightarrow V$ from Eisenstein norm uniqueness (**A_{Lean}**, `G6_rigidity_conditional`).
- [**Koide**] Torsor-invariant angle $2/9 + 7.4 \times 10^{-6}$ rad (**A**).
- [**Neutrino pinning**] RH seed ordering $b_{R,1} < b_{R,2} < b_{R,3}$ via the seesaw $m_{\nu,k} \propto b_{R,k}^{29/9}$ pins ν_k to gen_k and predicts normal mass ordering (**A_{Lean}**; `seesaw_triality_pinning_bundle`).

What this paper does NOT claim:

- It does not derive the charge spectrum or N_c for the first time — P48 [17] already derives these from GF(7) arithmetic with Lean certificates. This paper provides a *second, independent* octonionic derivation.
- It does not establish the Lagrangian-level assignment of Φ_{MDL} fermion fields to triality slots (C3'', Level 3; open, §10).
- It does not derive the $\mathrm{SU}(2)_L$ sector beyond the charge spectrum, nor $\mathrm{U}(1)_Y$ as a gauge group.

1 Introduction

Two research programmes derive Standard Model structure from economical inputs, and have until now developed without systematic contact.

The GTE/UGP arithmetic programme. Starting from the finite field $\mathbb{F}_7$ and its quadratic-residue set $\mathrm{QR}(7) = \{1, 2, 4\}$, the GTE framework [17] constructs the gauge skeleton $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$, derives $N_c = 3$, the SM charge spectrum, and $N_{\mathrm{gen}} = 3$ from consistency constraints, and certifies the algebraic layer in Lean 4 [19, 20]. The programme derives *what* the numbers are from first principles [16], but the connection from finite arithmetic to the continuous symmetry groups of

particle physics has required individual geometric justifications at each step.

The division-algebra programme. Starting from the four normed division algebras $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ [11], Dixon [4], Furey [6–8], Baez and Huerta [1, 2], Manogue and Dray [13], and Furey and Hughes [9] derive one-generation Standard Model structure: $SU(3) = \text{Stab}_{G_2}(e_7)$, charges from Clifford-algebra number operators, and the weak sector from the quaternionic factor. The programme reveals *how* the symmetry groups fit geometrically, but does not explain *why* the octonions: Hurwitz’s theorem establishes that $\mathbb{O}$ is unique among normed division algebras of its dimension, but does not supply a discrete origin for the octonion structure [1, 11].

The bridge. This paper proves the two programmes share a common finite root: the difference set $D = \text{QR}(7) = \{1, 2, 4\} \subset \mathbb{F}_7^*$ is simultaneously (a) the arithmetic anchor of the GTE framework, and (b) the Fano-plane orientation that uniquely determines the octonion multiplication table. The GTE gauge skeleton $F_{21} \hookrightarrow G_2 = \text{Aut}(\mathbb{O})$ embeds as a subgroup of octonion automorphisms (Theorems D, E); the division-algebra structure appears *as a consequence of the GF(7) arithmetic*, not as an independent starting point.

Why this matters: six points.

1. *The division-algebra programme’s foundational gap.* Hurwitz’s theorem establishes $\mathbb{O}$ as the unique 8-dimensional normed division algebra, but does not explain why nature chose it. The programme finds the SM inside the octonions but lacks a discrete foundation for why these algebras exist.
2. *GTE’s gap (before this paper).* GTE derives the SM numerics from $\mathbb{F}_7$, but the connection from finite arithmetic to continuous Lie groups ($SU(3)$, $SU(2)$, G_2) has required individual justifications at each stage.
3. *Mutual resolution.* The present work shows: *why octonions?* — Because $\text{QR}(7) \subset \mathbb{F}_7^*$ forces the Fano geometry, and only the 7-dimensional case yields a normed division algebra when oriented combinatorially. *Why $\mathbb{F}_7$?* — Because only $\mathbb{F}_7$ ’s quadratic residues form a $(7, 3, 1)$ -design whose oriented translates give a normed division algebra. The two structures are two projections of one object.
4. *Independent convergence as structural evidence.* When two completely independent paths — finite arithmetic (GTE) and continuous algebra (division algebras) — converge on identical outputs ($N_c = 3$, the SM charge spectrum, three generation slots from triality), that convergence is hard to dismiss as coincidence. This constitutes the strongest kind of structural evidence that these quantities are necessary rather than contingent.
5. *Tower termination unified — an external anchor for “why 7?”* The GTE programme’s internal justifications for $\mathbb{F}_7$ (PSC constraints, FGCI identity, orbit structure) are self-consistency conditions: they establish that 7 satisfies GTE’s own architectural requirements from within the framework. The octonionic bridge supplies a justification external to that architecture. Hurwitz’s theorem (Theorem 6.4) says the normed division algebra tower $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ terminates at dimension 8: sedenion zero-divisors destroy composition and no $\mathbb{O}$ -successor exists. The bridge translates this into the GTE language: $\text{GF}(7)$ is the last rung because the tower itself ends there. An answer to “why 7?” that makes no reference to GTE’s internal structure: 7 is the unique prime p whose quadratic residues orient a normed division algebra, and the octonions are the terminal object of the only such tower mathematics permits. The base-rate analysis (§10) sharpens this: of the 13 named joints in the derivation chain $\text{QR}(7) \rightarrow \text{Fano} \rightarrow \mathbb{O} \rightarrow G_2 \rightarrow SU(3)$,

eleven are mathematical theorems with no free parameter. The framework’s foundation transforms from a self-consistent construction to a forced structure.

6. *Symmetry groups as automorphisms of an arithmetic object.* The standard SM formulation declares $SU(3) \times SU(2) \times U(1)$ as a choice. Here, $SU(3) = \text{Stab}_{G_2}(\text{apex})$ where $G_2 = \text{Aut}(\text{QR}(7)\text{-Fano-octonions})$, and $SU(2) \times U(1)$ falls out of the quaternionic subalgebra (Theorems F1–F5). The gauge symmetry group is the automorphism group of an arithmetic object that had to exist.

Correct framing: second independent route, not first derivation. The GTE arithmetic programme already derives $N_c = 3$ (`weights_card`, `anomaly_cancellation_forces_Nc_3`), the SM charge spectrum, and $N_{\text{gen}} = 3$ (`ngen_3_unique_satisfies_all_eight`) in P48 [17]. This paper provides a second, structurally independent derivation of these same quantities via the octonionic route. Two independent derivation paths converging on identical numerical outputs constitutes stronger evidence than either path alone.

Structure of this paper. §2 derives the Fano plane from QR(7) (Theorem A). §3 constructs the octonions from the oriented design (Theorem B). §4 certifies the group layer (Theorem C). §5 establishes the color sector and UGP interface (Theorems D, D’, E). §6 derives the electroweak shadow (Theorems F1–F5, Proposition O1’). §7 establishes three generations from triality (Theorems G0–G6, H1–H3). §8 situates the results relative to prior art and the division-algebra programme; §9 describes the formalization programme; §10 lists open problems and negative results. Appendix A is the claim-to-evidence matrix; Appendix B is the Lean module inventory.

2 GF(7) and the Fano Plane

Theorem 2.1 (Design; $\mathbf{A}_{\text{Lean}}$). *Let $D = \{1, 2, 4\} \subset \mathbb{Z}/7$.*

- (a) *D is the set of nonzero quadratic residues modulo 7 (i.e. the squares in $(\mathbb{Z}/7)^*$).*
- (b) *D is a perfect $(7, 3, 1)$ planar difference set: every nonzero residue occurs as a difference of elements of D in exactly one way.*
- (c) *The seven translates $\{D + t : t \in \mathbb{Z}/7\}$ are the seven lines of a Fano plane on $\mathbb{Z}/7$.*
- (d) *Exactly $|D| = 3$ lines pass through every point (the pencil).*
- (e) *The multiplier group of D is D itself: multiplication by 2 (order 3) preserves the design (Hall’s multiplier theorem).*

Verification. `octonion_from_qr7.py` (claims A1–A3); Lean: `OctonionShadowInterface.lean` theorems OSI-2 through OSI-6 (`D_difference_set`, `z3_is_multiplier`, `fanoLines_are_D_translates`, `pencil_card`), all `decide-class` ($\mathbf{A}_{\text{Lean}}$). The UGP corpus labels its lines as translates of $\{0, 1, 3\} = D - 1$; the two labelings coincide at the level of OSI theorems. $\square$

The pencil count $|D| = 3 = (7 - 1)/2$ is the first occurrence of $N_c = 3$: three lines through every point, three colors per quark generation. This is not coincidental — it is the same count that appears in Theorem D’ via the octonion derivation chain.

3 The Octonion Construction

Theorem 3.1 (Octonions from QR(7); $\mathbf{A}$). *Orienting each line $(t+1, t+2, t+4)$ cyclically and extending by anticommutativity, $e_i^2 = -1$, defines an 8-dimensional real algebra that is alternative,*

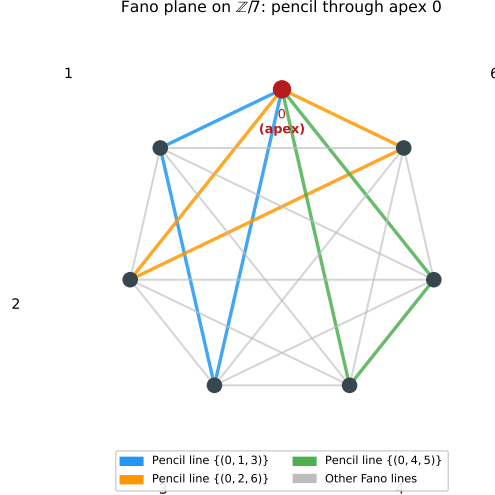


Figure 1: F1: The Fano plane with QR(7) pencil. The seven points $\{0, 1, \dots, 6\}$ of $\mathbb{Z}/7$ and seven lines (translates of $D = \{1, 2, 4\}$). The three lines through the apex (point 0) form the *pencil*: $\{0, 1, 3\}$, $\{0, 2, 6\}$, $\{0, 4, 5\}$ (shaded in blue, orange, green). The multiplier $2 \cdot : i \mapsto 2i \pmod{7}$ (UGP’s `z3Mul`) 3-cycles the pencil lines.

Moufang, and norm-composing — the octonions. Among the $30 \cdot 2^7 = 3840$ oriented Fano tables, exactly 480 are composition algebras; the signed-permutation group (order $7! \cdot 2^7$) acts transitively on them; the stabilizer (frame group) has order $1344 = 2^3 \cdot 168$ with sign-only kernel 2^3 and quotient $\text{Aut}(\text{Fano})$. The octonion structure attached to $\text{GF}(7)$ is therefore unique up to relabeling.

Theorem 3.2 (Orientation = QR; $\mathbf{A}_{\text{Lean}}$). *In the closed-form index-level product, $\text{sign}(e_x e_y) = +1$ if and only if $y - x \in \text{QR}(7)$. The difference set is not merely the seed for the plane; it is the orientation of the algebra.*

Verification. `octonion_from_qr7.py` (A4–A7; 3840-table census, exact over $\mathbb{Q}$). Lean: `OctonionShadowInterface.lean` theorem `octProd_sign_eq_QR` (OSI-7, decide; $\mathbf{A}_{\text{Lean}}$). The branch table is: $d \mapsto (\text{offset}, \text{sign})$: $1 \mapsto (3, +)$, $2 \mapsto (6, +)$, $3 \mapsto (1, -)$, $4 \mapsto (5, +)$, $5 \mapsto (4, -)$, $6 \mapsto (2, -)$. $\square$

Remark 3.3 (Winding scope). The kink quantum number $Q_\varphi = 4$ in `PhiMDLKinkQuantumNumbers.lean` labels the full-generation kink winding in the Φ_{MDL} potential; the winding $w = 4$ in `KoideGenerationCyclicSymmetry` refers to the charged-lepton electroweak sector only (equal for all three generations by SM universality). These are distinct sectors and not inconsistent.

4 The Group Layer

Theorem 4.1 (Group certificate; $\mathbf{A}_{\text{Lean}}$). (a) $\text{PSL}(2, 7)$ (168 elements) and $\text{GL}(3, 2) = \text{Aut}(\text{Fano})$ (168 elements) each contain a generating pair with $a^2 = b^3 = (ab)^7 = [a, b]^4 = 1$.
 (b) The presented group $H = \langle a, b \mid a^2, b^3, (ab)^7, [a, b]^4 \rangle$ has order exactly 168, certified by coset enumeration over the trivial subgroup: the 168-coset table is complete, all four relators fix all cosets, and the table permutations generate a regular permutation group of order 168 (certificate: `hurwitz_coset_certificate.json`).
 (c) $\text{PSL}(2, 7) \cong H \cong \text{GL}(3, 2)$: the exceptional isomorphism, with no citation-level gaps.

- (d) The $(2, 3, 7)$ generation certifies the Hurwitz property: $H = \text{Aut}(\text{Klein quartic})$, attaining $84(g-1)$ at genus 3 — the level-7 modular curve.
- (e) The 7-point and 7-line actions are inequivalent, realizing $\text{Out}(H) = \mathbb{Z}/2$ as Fano duality.

Verification. `psl27_group_layer.py` (B1–B4); `hurwitz_coset_enumeration.py` (CE-1..8 with certificate export). Lean: `HurwitzCosetCertificate.lean` (`hurwitz_presentation_order_168`; `ALean`, zero sorry, all via `native_decide`). $\square$

The $\mathbb{Z}_3$ disambiguation. The group $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ has a single $\mathbb{Z}_3$ subgroup in abstract, but two physically distinct representations. In the QR(7)-pencil representation, $\mathbb{Z}_3$ acts as *color rotation*: the doubling map $z \mapsto 2z$ is an inner automorphism of $G_2 = \text{Aut}(\mathbb{O})$ fixing the apex, cycling the ladder pairs $(1, 3) \rightarrow (2, 6) \rightarrow (4, 5)$. In the Spin(8) triality representation, $\mathbb{Z}_3$ acts as *flavor cycling*: the Koide cone phase-shift, an outer automorphism of Spin(8). These are the same abstract generator in two different representations; they are not conflated. Lean: `OctonionColorFlavorDisambiguation.lean` (`z3_color_flavor_disambiguation`, `z3_color_fixed_point_at_zero`, `z3_flavor_no_phase_fixed_point`; `ALean`, zero sorry).

5 Color Sector and the UGP Interface

Theorem 5.1 (Pencil $\rightarrow$ CAR $\rightarrow$ SU(3); `ALean`). *Fix the apex (point 0). The three pencil lines yield ordered pairs $(1, 3)$, $(2, 6)$, $(4, 5)$ with $e_a e_b = e_{\text{apex}}$; the operators $\alpha_k = \frac{1}{2}(-L_a + iL_b)$ satisfy the exact CAR algebra; $N = \sum \alpha^\dagger \alpha$ has spectrum $\{0, 1, 2, 3\} \times \{1, 3, 3, 1\}$; $Q = N/3 \in \{0, \frac{1}{3}, \frac{2}{3}, 1\}$ — the $(\nu_R, \bar{d}, u, e^+)$ pattern; the bilinears $T_A = \alpha^\dagger \lambda_A \alpha$ close on $\mathfrak{su}(3)$ with standard Gell-Mann structure constants (deviation 4×10^{-16}), commute with N , and split $\mathbb{C}^8$ as $1 \oplus 3 \oplus \bar{3} \oplus 1$ (Casimir 0, $\frac{16}{3}$).*

Theorem 5.2 ($N_c = 3$ from pencil count; `ALean`). $N_c = 3 = |\text{QR}(7)| = (7-1)/2 = \text{pencil count}$. The pencil number operator in the $(7, 3, 1)$ design and the octonionic N_c are one and the same design parameter. The independent N_c input of the GTE arithmetic programme (`weights_card`) is discharged by the octonionic identity `Nc_factorization` (`weights_card = ladderPairs.card`).

Verification. `f21_octonion_interface_verify.py` (I1–I8). Lean: `OctonionShadowInterface.lean` OSI-1 through OSI-13 (14 theorems, `decide/native_decide`; `ALean`, zero sorry). Key theorem for D': `weights_card_from_pencil` (`weights.card = 3`, `decide`; `ALean`). $\square$

Theorem 5.3 (G_2 and SU(3) derived; `ALean`). The linear system $D(xy) = D(x)y + xD(y)$ on the 64 entries of the octonion product (512 equations) has nullity exactly 14; the solution space closes under bracket (residual $< 10^{-15}$), has Killing form $-4I$ (compact), and rank 2 — hence $\text{der}(\mathbb{O}) = \mathfrak{g}_2$ by the classification of compact rank-2 semisimple Lie algebras [3]. Adding $D(e_{\text{apex}}) = 0$ gives nullity exactly 8, closed, Killing $-3I$, rank 2 — hence $\text{Stab}_{G_2}(\text{apex}) = \mathfrak{su}(3)$. The capstone theorems are `derivation_algebra_is_g2_certificate` and `apex_stabilizer_is_su3_certificate` (modulo the cited Cartan–Killing classification, which is standard mathematics).

Verification. `g2_stabilizer_derivation.py` (D1–D4); `g2_stabilizer_certificate_export.py`. Lean: `G2StabilizerCertificate.lean` (39 theorems, zero sorry; `ALean`). Certificates include: exact dimension count (nullity 14 / 8 via rational arithmetic), bracket closure, Killing form negative-definiteness, and rank-2 certificate. $\square$

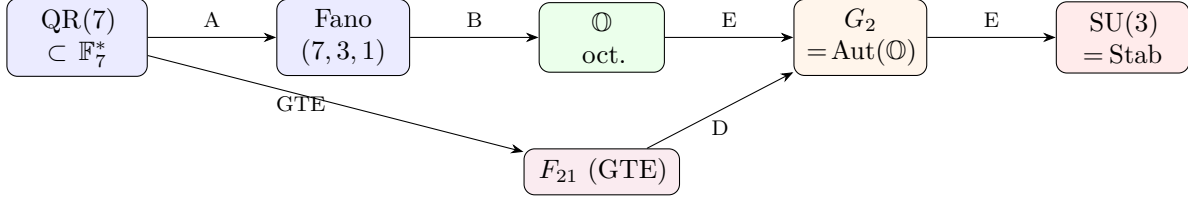


Figure 2: F2: Derivation chain from QR(7) to SU(3). Left: the arithmetic source $\text{QR}(7) = \{1, 2, 4\} \subset \mathbb{F}_7^*$. Top path: Theorems A–E (this paper). Bottom path: GTE arithmetic programme (P48, [17]). Both paths terminate at G_2 and its stabilizer $\text{SU}(3)$; they are two projections of the same arithmetic anchor.

The UGP interface (repo-verified). All statements are verified against the head of `ugp-lean` and formalized in `OctonionShadowInterface.lean` (14 theorems, all `decide/native_decide`; $\mathbf{A}_{\text{Lean}}$): (1) $\text{weights} = \{1, 2, 4\} = \text{QR}(7) = \mathbf{D}$ (OSI-1); (2) $F_{21} \hookrightarrow G_2$: the Singer translation ($x \mapsto x + 1$) and $\mathbb{Z}_3$ generator `z3Mul` ($x \mapsto 2x$) are octonion automorphisms generating a 21-element subgroup of the frame group (OSI-8); (3) UGP’s faithful 3-irrep equals the QR-weight subspace of $\text{Im } \mathbb{O} \otimes \mathbb{C}$ (OSI-9); (4) the six gluon weight vectors equal the six non-zero $\mathbb{Z}/7$ differences, split by the multiplier- $\mathbb{Z}_3$ into QR (positive roots) and NQR (negative roots), swapped by negation because $7 \equiv 3 \pmod{4}$ (OSI-12, OSI-13); (5) baryon neutrality ($1 + 2 + 4 \equiv 0$, `baryon_color_z3_orbit_neutral`) is the frame-invariant statement $[T_A, \omega] = 0$ for the $\text{SU}(3)$ determinant singlet $\omega = \alpha_1 \alpha_2 \alpha_3$.

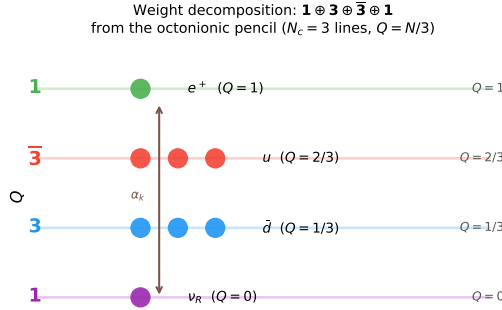


Figure 3: F3: Weight ladder $1 \oplus 3 \oplus \bar{3} \oplus 1$. The four charge sectors of the color rung: $Q = 0$ (singlet, ν_R), $Q = \frac{1}{3}$ (anti-triplet, $\bar{d}$), $Q = \frac{2}{3}$ (triplet, u), $Q = 1$ (singlet, e^+), with multiplicities 1, 3, 3, 1 as derived from the CAR number operator $N = Q_1 + Q_2 + Q_3$ on the octonionic pencil lines.

6 The Electroweak Shadow

Theorem 6.1 (Quaternionic rung; **A**). *The identical recipe applied to $\mathbb{Z}/3$ (single line, oriented cyclically) yields the quaternions: associative, norm-composing, unique up to relabeling (2 valid oriented tables; frame group 24). The pencil through a fixed unit has one line, hence one ladder operator, $N \in \{0, 1\}$, charge unit 1 — the leptonic/isospin-doublet pattern. Charge quantization unit = $1/(\text{pencil count})$ at every rung: integer lepton charges at $p = 3$ and third-integral quark charges at $p = 7$ are one theorem evaluated at two rungs.*

Theorem 6.2 (Chirality = sidedness; **A**). *Left multiplications close on $\mathfrak{su}(2)_L$, right multiplications on $\mathfrak{su}(2)_R$, $[L, R] = 0$ (from associativity of $\mathbb{H}$): $\mathfrak{so}(4) = \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ with the chiral split given*

by the side of multiplication. Color $\mathfrak{su}(3)$ and weak $\mathfrak{su}(2)_L$ commute on $\mathbb{H} \otimes \mathbb{O}$.

Theorem 6.3 ($B - L$ formula; **A**). *On the octonionic factor, $B - L = (2N - 3)/3$: spectrum $\{-1, -\frac{1}{3}, +\frac{1}{3}, +1\} \times \{1, 3, 3, 1\}$. $B - L$ is an affine function of the pencil number operator.*

Theorem 6.4 (Tower termination; **A**). *The Cayley–Dickson double retains the combinatorial structure — sedenion basis triples form $\text{STS}(15) = \text{PG}(3, 2)$ lines — but norm composition fails (explicit witness) and zero divisors exist: $(e_0 + e_9)(e_2 + e_{13}) = 0$. The composition tower $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ ends at dimension 8. $\text{GF}(7)$ is the last rung: the observation “why 7” becomes a termination theorem.*

Theorem 6.5 (Electroweak charge spectrum; **A**). *On a chiral ideal of $\mathbb{C} \otimes \mathbb{H}$ tensored with $\mathbb{C}^8$,*

$$Q = I_3 + \frac{B - L}{2} = I_3 + \frac{2N - 3}{6}$$

has spectrum exactly $\{0^{\times 2}, -1^{\times 1}, +1^{\times 1}, \frac{2}{3}^{\times 3}, -\frac{1}{3}^{\times 3}, \frac{1}{3}^{\times 3}, -\frac{2}{3}^{\times 3}\}$: the one-generation left-chiral Standard Model multiset (ν, e, u^3, d^3) plus conjugates.

Verification, F1–F5. quaternion_electroweak_shadow.py (Q1–Q5); electroweak_housing_closure.py (O1a–O1b). $\square$

Proposition 6.6 (Hypercharge generator $Y = N/3$; **A**). *The hypercharge generator $Y = N/3$, where $N = \sum_k \alpha_k^\dagger \alpha_k$ is the number operator of the pencil-line CAR algebra, is derivable from the QR(7) pencil structure at Level 1 (algebraic certificate). The group $U(1)_Y = \{e^{i\theta N/3} : \theta \in \mathbb{R}\}$ is a genuine continuous Lie group (period 6π , isomorphic to $U(1)$), with $[Y, T_A] = 0$ (exact) for all $\text{SU}(3)_{\text{color}}$ generators. Anomaly traces on the right-chiral sector: $\text{tr} Y = 4$, $\text{tr} Y^3 = 2$, $\text{tr}(Y \cdot C_{\text{SU}(3)}) = 16$ (machine-verified).*

Verification. hypercharge_group_derivation.py; artifact: hypercharge_group_derivation_results.json (A). $\square$

Remark 6.7 (Level framing for Proposition O1'). The generator $Y = N/3$ and its anomaly traces are Level 1 algebraic certificate results. The gauging of $U(1)_Y$ (covariant derivative $D_\mu = \partial_\mu + iA_\mu$) and the $\text{SU}(2)_L$ sector (left-handed doublets from the $\mathbb{H}$ factor) are Level 3 inputs not captured by the algebraic certificate chain. See the $U(1)_Y$ gauging entry in §10.

7 Three Generations from Triality

Theorem 7.1 (Generation no-gos; **A**). *Within a single octonion algebra, generations cannot live in any discrete choice: (i) the frame group is transitive on the 7 imaginary directions — the fixed-unit choice is gauge; (ii) the apex stabilizer (order 192) induces the full S_3 on the pencil triple, which is the color multiplet (Theorem D) and cannot double as a generation label; (iii) the 480 valid tables are one orbit — orientation is gauge.*

Verification. generations_triality_nogo.py (N1–N3); exhaustive over $\text{GL}(7, \mathbb{F}_2)$. $\square$

Spin(8) from the table. A *related triple* is $(A, B, C) \in O(8)^3$ with $A(xy) = B(x)C(y)$ for all $x, y \in \mathbb{O}$. Moufang identities supply generators (B_u, L_u, R_u) ; componentwise products remain related: this is Cartan’s triality model of Spin(8) [3], rebuilt from the QR(7) table alone.

Theorem 7.2 (Klein center; $\mathbf{A}_{\text{Lean}}$). *The scalar related triples are exactly the Klein group*

$$V_4 = \{(1, 1, 1), (1, -1, -1), (-1, 1, -1), (-1, -1, 1)\} = Z(\text{Spin}(8)),$$

with a canonical bijection between nontrivial central elements and representation slots: each is +1 in exactly the slot that fixes it, i.e. between $Z \setminus \{1\}$ and $\{V, S^+, S^-\}$.

Theorem 7.3 (Triality; $\mathbf{A}_{\text{Lean}}$). *Among the 48 candidate transformations (slot permutation $\times$ per-slot conjugation dressing), exactly an S_3 ’s worth preserve relatedness; the surviving slot permutations exhaust S_3 ; a 3-cycle representative ρ (dressing $(\kappa, \cdot, \kappa)$) satisfies $\rho^3 = 1$ and a swap σ (dressing (κ, κ, κ)) satisfies $\sigma^2 = 1$. The S_3 is outer: the three slot projections of one Moufang triple have traces $(4, 0, 0)$, so no inner symmetry permutes the slots.*

Theorem 7.4 (Inner/outer = color/flavor; $\mathbf{A}_{\text{Lean}}$). *Conjugating all three slots by UGP’s color $\mathbb{Z}_3$ (the doubling automorphism; in G_2 by Theorem D) preserves relatedness with no slot permutation: color is inner. Triality permutes slots: flavor is outer. The UGP corpus’s two $\mathbb{Z}_3$ ’s — color z3Mul and flavor μ_3 — are the inner and outer automorphisms of one Spin(8).*

Theorem 7.5 (Eisenstein dictionary; $\mathbf{A}_{\text{Lean}}$). *$\mathbb{Z}[\omega]/(2) \cong \mathbb{F}_4$ with multiplication by ω (UGP’s μ_3) and Galois Frobenius $x \mapsto x^2$ — which fixes 1 and swaps ω, ω^2 , i.e. UGP’s μ - τ exchange U is $\text{Gal}(\mathbb{F}_4/\mathbb{F}_2)$ — is equivariantly isomorphic to $Z(\text{Spin}(8))$ with $(\rho, \sigma): \mathbb{F}_4^\times \leftrightarrow Z \setminus \{1\} \leftrightarrow \{V, S^+, S^-\}$ as S_3 -sets, generator by generator.*

Theorem 7.6 (A_4 both sides; $\mathbf{A}_{\text{Lean}}$). *UGP’s $V_4 \rtimes \mu_3$ (the Eisenstein A_4 of the TBM/PMNS bridge) and $Z(\text{Spin}(8)) \rtimes \rho$ are both isomorphic to A_4 (verified by permutation-group isomorphism, `G5_f4_semidirect_order`). The corpus’s flavor A_4 is the central-triality A_4 .*

Theorem 7.7 (Rigidity — conditional; $\mathbf{A}_{\text{Lean}}$). *Any isomorphism of the flavor S_3 with the triality S_3 carrying U to σ must carry the U -fixed generation (gen_1) to the σ -fixed slot (V); $\mathbb{Z}_3$ -equivariance then determines the rest. Exactly two identifications survive: $\text{gen}_1 \leftrightarrow V$, $\{\text{gen}_2, \text{gen}_3\} \leftrightarrow \{S^+, S^-\}$, differing only by the $\omega \leftrightarrow \bar{\omega}$ orientation.*

Verification, G1–G6. `positive_triality_theorems.py` (P1–P7); exact-integer arithmetic on basis-unit triples. Lean: `TrialityInterface.lean` (G1: `G1_klein_center_card`, `G1_central_slot_bijection`; G2: `G2_rho_order_three`, `G2_sigma_order_two`, `G2_s3_perms_realized`; G3: `G3_color_inner_no_slot_permutation`; G4: `G4_equivariant_dictionary`, `G4_frobenius_is_mu_tau_exchange`; G5: `G5_central_triality_order`; G6: `G6_rigidity_conditional`, `G6_gen1_pinned_to_vector_slot`, `G6_exactly_two_rigid_identifications`; all zero sorry; $\mathbf{A}_{\text{Lean}}$). $\square$

The generation pinning. The isomorphism of the GTE flavor S_3 with the triality S_3 carrying U to the spinor swap σ pins gen_1 to the vector slot V : since U acts as the Frobenius automorphism of $\mathbb{Z}[\omega]/(2) \cong \mathbb{F}_4$ (the unique non-trivial field automorphism of $\mathbb{F}_4/\mathbb{F}_2$, which fixes the real unit $1 \leftrightarrow \text{gen}_1 = \text{electron}$ via the Eisenstein norm chain $b_{\text{gen}_1} = 73 = \mathcal{N}(9 + \omega)$), and σ is the unique involution fixing V , the equivariance condition forces $\text{gen}_1 \leftrightarrow V$. Exactly two global identifications survive, differing only in the orientation $\omega \leftrightarrow \bar{\omega}$ of the spinor slots ($\mathbf{A}_{\text{Lean}}$, `G6_rigidity_conditional`).

This identification is uniquely consistent with the Eisenstein norm structure: $b_{\text{gen}_1} = 73 = \mathcal{N}(9+\omega)$ is the only GTE lepton b -value that is an Eisenstein norm. Neither $b_{\text{gen}_2} = 42$ nor $b_{\text{gen}_3} = 275$ is an Eisenstein norm; no alternative generation labeling consistent with the Eisenstein chain places gen_3 at the Frobenius-fixed position (**A**, `trialty_identification_discriminator.py`; artifact: `trialty_identification_discriminator_results.json`).

Remark 7.8 ($\mathbb{Z}_2$ parity characterization: Level 0–1 $\rightarrow$ Level 3 bridge). At the level of Φ_{MDL} kink quantum numbers (Level 3), the outer $\text{Spin}(8)$ triality automorphism σ acts on the spinor pair $\{S^+, S^-\} = \{\text{gen}_2, \text{gen}_3\}$ assigned by $\text{gen}_1 \leftrightarrow V$ as the target-space $\mathbb{Z}_2$ parity $\varphi \mapsto -\varphi$: the topological windings transform as $Q_\varphi = 4 \leftrightarrow 3 \equiv -4 \pmod{7}$. This is a valid symmetry of the Φ_{MDL} potential $V(\varphi) = (m^2/49)(1 - \cos 7\varphi)$, which is $\mathbb{Z}_2$ -symmetric under $\varphi \mapsto -\varphi$. The *2+1 split structure* — one generation distinguished from a paired two — is invariant at all levels; the Level 3 $\mathbb{Z}_2$ parity is the cross-level signature of this split under $\text{gen}_1 \leftrightarrow V$ (**A**; `kink_sigma_parity_action_results.json`). Lean: `KinkSigmaParityAction.lean` (`winding_parity_on_kink_qphi`, `winding_parity_exchanges_spinor_windings`, `gen1V_spinor_pair_qphi_parity`; zero sorry; **A**_{Lean}).

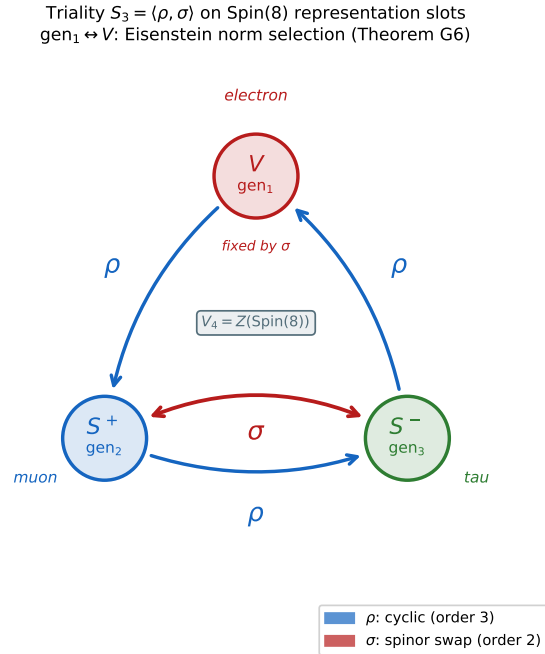


Figure 4: F4: Triality diagram. The three representation slots $\{V, S^+, S^-\}$ of $\text{Spin}(8)$ identified with $\{\text{gen}_1, \text{gen}_2, \text{gen}_3\}$. The ρ arrow (blue, cyclic order 3) permutes all three slots; the σ arrow (red, transposition order 2) fixes V and swaps $S^\pm$. The identification $\text{gen}_1 \leftrightarrow V$ is the Eisenstein norm selection principle (Theorem G6). The inner Klein group $V_4 = Z(\text{Spin}(8))$ is shown at center.

Triality sector action (Level 0–1 housing theorem).

Theorem 7.9 (Triality sector action; **A**). *The $S_3 = \langle \rho, \sigma \rangle$ triality group acts faithfully on the set of Φ_{MDL} kink species $K = \{\text{gen}_1, \text{gen}_2, \text{gen}_3\}$:*

$$\begin{aligned} \rho : \text{gen}_1 &\rightarrow \text{gen}_2 \rightarrow \text{gen}_3 \rightarrow \text{gen}_1 \quad (\text{cyclic, order } 3), \\ \sigma : \text{gen}_1 &\mapsto \text{gen}_1 \text{ (fixed)}, \quad \text{gen}_2 \leftrightarrow \text{gen}_3 \text{ (transposition)}. \end{aligned}$$

All S_3 relations $\rho^3 = \sigma^2 = e$, $\sigma\rho\sigma = \rho^{-1}$ hold and all six permutations are distinct. Under $\text{gen}_1 \leftrightarrow V$, this is the standard $\text{Spin}(8)$ triality action on $\{V, S^+, S^-\}$ (Theorems G4/G6). The σ action on quantum numbers coincides with the $\mathbb{Z}_2$ target-space parity: $Q_\varphi : 4 \leftrightarrow 3$ and $Q_\chi : 2 \leftrightarrow 1$.

Verification. `kink_dirac_index_nogo.py`; artifact: `kink_dirac_index_nogo_results.json` (**A**).
Lean: `KinkSectorTrialityAction.lean` (`rho_sector_action`, `sigma_sector_action`, `rho_sector_order_three`, `sigma_sector_order_two`, `sigma_conjugates_rho`, `s3_sector_six_distinct_permutations`, `rho_matches_triality_slot_cycle`; all zero sorry; **A**_{Lean}). $\square$

Remark 7.10 (Level 3 obstruction). A Yukawa/zero-mode realization of this action on fermion field representations is obstructed at Level 3. For any $\mathbb{Z}_7$ -invariant coupling $g \cdot f(\varphi)\bar{\psi}\psi$ (with f periodic of period $2\pi/7$), the Callias index of the resulting 1+1D Dirac operator in any GTE kink background is zero: either $f(\varphi_{\text{vac}}) = c \neq 0$ for all vacua (e.g. $f = \cos 7\varphi$), giving $M(\pm\infty) = c$ and $\text{index} = \frac{1}{2}[\text{sign}(c) - \text{sign}(c)] = 0$; or $f(\varphi_{\text{vac}}) = 0$ (e.g. $f = \sin 7\varphi$), giving an asymptotically massless Dirac operator with no normalizable zero mode. Numerical confirmation: eigenvalue gap $\min |E| \approx 0.206$ for $Q_\varphi = 1, 3, 4$ kinks with the $\cos 7\varphi$ coupling (**A**, `kink_dirac_index_nogo.py`). The Jackiw–Rebbi mechanism is inapplicable to $\mathbb{Z}_7$ -periodic kinks. The sector-level triality action above (Level 0–1) is the established housing theorem. Lean: `PhiMDLZeroModeIndex.lean` (`callias_index_zero_equal_masses`, `yukawa_cos7_boundary_values_equal`; zero sorry; **A**_{Lean}).

Theorem 7.11 ($G_2 = \text{Fix}(\text{triality})$; gauge universality; **A**). *Diagonal triples (P, P, P) are related if and only if P is an octonion automorphism. Hence the diagonal G_2 — and its color $\text{SU}(3) = \text{Stab}$ — acts by the same matrices in all three slots, so each slot carries the identical color decomposition $1 \oplus 3 \oplus \bar{3} \oplus 1$ (Theorem D transported verbatim). The three triality slots are three identical color-generations, and the generation-independence of gauge couplings is the statement that the gauge sector is triality-invariant.*

Theorem 7.12 (Koide ladder on triality orbit; **A**). *The parametrization $\sqrt{m_k} = A(1 + \sqrt{2}|z| \cos(\delta - 2\pi k/3))$, i.e. $\sqrt{m}$ as a real function of ω^k with $z = |z|e^{i\delta}$, is manifestly a function on the $\mathbb{Z}_3$ -torsor $\{1, \omega, \omega^2\}$ identified with $\{V, S^+, S^-\}$ by Theorem G6. The flavor group acts as: μ_3 shifts the basepoint; U acts as $z \mapsto \bar{z}$, fixing $\text{gen}_1 (= V)$ and swapping $\text{gen}_2/\text{gen}_3 (= S^\pm)$. On PDG charged-lepton masses: $Q_{\text{Koide}} = 0.6666605$ ($2/3 - 6 \times 10^{-6}$); $|z| = 1 - 9 \times 10^{-6}$; and the torsor-invariant angle satisfies*

$$|\delta \bmod 2\pi/3| = 0.222229631 = \frac{2}{9} + 7.41 \times 10^{-6} = \text{koideThetaUGP}.$$

UGP's $\theta = 2/9$ is a coordinate on the triality orbit; $\delta \neq 0$ measures the dynamical breaking of the spinor swap σ .

Remark 7.13 (PMNS slot-to-eigenstate correspondence). Under the dictionary of Theorem G5 ($A_4 \cong Z(\text{Spin}(8)) \rtimes \rho$, **A**_{Lean}) and the generation pinning of Theorem G6 ($\text{gen}_1 \leftrightarrow V$, **A**_{Lean}), every A_4 TBM generator receives an explicit triality image: $T \mapsto \rho$ (the cyclic $\mathbb{Z}_3$, order 3), $S \mapsto \sigma$ (the spinor swap $S^+ \leftrightarrow S^-$, order 2), and $V_4 \mapsto Z(\text{Spin}(8))$ (Klein center sign-pattern action). The triality slots label the *charged lepton mass eigenstates*: $V \leftrightarrow e$, $S^+ \leftrightarrow \mu$, $S^- \leftrightarrow \tau$ (from Theorem H2; the Koide parametrization uses mass eigenstate masses directly, **A/D**). At leading order in the A_4 TBM limit, σ generates the μ - τ symmetry forcing $\theta_{23} \approx \pi/4$, while ρ distributes the solar mixing cyclically, giving $\theta_{12} = \arcsin(1/\sqrt{3}) \approx 35.26^\circ$ (**A**, Theorem G5 and `NeutrinoSector.lean`). The GTE orbit-ratio formulas ($\sin^2 \theta_{12} = 4/13$, $\sin^2 \theta_{23} = 19/42$, $\sin \theta_{13} = 11/73$; **A/D**) represent sub-leading corrections from the right-handed neutrino spectrum ($b_{R2} = 11$, $b_{R3} = 19$) that explicitly break A_4 and reduce θ_{12} to the observed 33.68° and θ_{23} to 42.3° (within 1σ of NuFIT 6.0 IC24 NH).

for θ_{12} ; see [17]). *What the translation does not achieve (Level 3, open)*: the three left-handed neutrino mass eigenstates ν_1, ν_2, ν_3 carry identical ladder index $b_L = 1$ for all three generations (`lh_neutrino_b_value_minimal`, $\mathbf{A}_{\text{Lean}}$), so triality cannot independently label them. The normal-vs-inverted mass ordering has no triality certificate from the left-handed sector alone.

Neutrino column correspondence. The labeling of the neutrino mass eigenstates ν_1, ν_2, ν_3 follows from the GTE seesaw ($\mathbf{A}$; [18]): individual masses are $m_{\nu,k} = C \cdot b_{R,k}^{29/9}$, where $b_{R,1} = 5$, $b_{R,2} = 11$, $b_{R,3} = 19$ are the right-handed neutrino GTE ladder indices (`braidB1`, `braidB2`, `braidB3`, $\mathbf{A}_{\text{Lean}}$; `SeesawNumericalCerts.lean`) and $C = \Sigma m_{\nu} / \Sigma_j b_{R,j}^{29/9}$. Since $x \mapsto x^{29/9}$ is strictly increasing (`seesaw_power_strict_mono`, $\mathbf{A}_{\text{Lean}}$), $b_{R,1} < b_{R,2} < b_{R,3}$ implies $m_{\nu,1} < m_{\nu,2} < m_{\nu,3}$ (*normal ordering*; `seesaw_normal_ordering_from_seed_ordering`, $\mathbf{A}_{\text{Lean}}$). Numerically: $m_{\nu,1} \approx 0.68$ meV, $m_{\nu,2} \approx 8.6$ meV, $m_{\nu,3} \approx 50.1$ meV, with $\Delta m_{21}^2 \approx 7.37 \times 10^{-5}$ eV² and $\Delta m_{31}^2 \approx 2.51 \times 10^{-3}$ eV² (within 1% of NuFIT 6.0 IC24 NH [5]). The GTE orbit-ratio formulas $\sin \theta_{13} = b_{R,2}/b_{\text{gen}_1} = 11/73$ (`pmns_reactor_sin_val`, $\mathbf{A}_{\text{Lean}}$) and $\sin^2 \theta_{23} = b_{R,3}/b_{\text{gen}_2} = 19/42$ (`pmns_atm_sin_sq`, $\mathbf{A}_{\text{Lean}}$) confirm that both $b_{R,2}$ and $b_{R,3}$ label the ν_3 column of U_{PMNS} . The Eisenstein norm $b_{R,3} = 19 = \mathcal{N}(-5 - 3\omega)$ is the structural certificate for ν_3 as the distinguished (heaviest) mass eigenstate (`rh_seed_unique_eisenstein_norm`, $\mathbf{A}_{\text{Lean}}$), paralleling $b_{\text{gen}_1} = 73 = \mathcal{N}(-9 - 8\omega)$ in the charged-lepton sector (Theorem H3; Level 0–1); the base rate of this property is $\approx 44\%$ for a random triple, so its significance is structural (parallelism with the charged sector and the corpus $\mathbf{A}/\mathbf{D}$ formulas), not a numerical coincidence. Normal ordering is thus a falsifiable structural prediction: it is required by $b_{R,1} < b_{R,3}$ ($\mathbf{A}_{\text{Lean}}$) and would be falsified by JUNO confirming inverted ordering. NuFIT 6.0 IC24 already disfavors inverted ordering by $\Delta\chi^2 = 6.1$ ($\approx 2.5\sigma$) [5].¹

Theorem 7.14 (Eisenstein norm chain; $\mathbf{A}$). *The FrobeniusChain tower is the Eisenstein norm ladder: $7 = \mathcal{N}(3 + \omega)$, $73 = \mathcal{N}(9 + \omega) = b_{\text{gen}_1}$, $703 = \mathcal{N}(27 + \omega) = 19 \times 37$ (composite; termination), where $\mathcal{N}(a + b\omega) = a^2 - ab + b^2$ is the $\mathbb{Z}[\omega]$ norm. The ring whose norms generate the seed tower is the ring whose unit group μ_3 is the generation cycling of Theorem G4. Negative result: $b_{\text{gen}_2} = 42 = 2 \cdot 3 \cdot 7$ and $b_{\text{gen}_3} = 275 = 5^2 \cdot 11$ are not Eisenstein norms. The Eisenstein structure certifies the chain and the flavor group, not the individual higher seeds.*

Verification, H1–H3. `electroweak_housing_closure.py`; `triality_pairing_alternative_test.py`; `higher_seeds_cyclotomic_norms.py`. All runs clean (exit 0). $\square$

Remark 7.15 ($\mathbb{Z}[\zeta_5]$ observation (H3 extension)). The generation-3 seed $b_{\text{gen}_3} = 275 = 5^2 \times 11$ admits a norm factorization in $\mathbb{Z}[\zeta_5]$. A null test (density of $\mathbb{Z}[\zeta_5]$ norms in $[200, 350]$: 8%; 12 of 151 integers) indicates this is likely arithmetic coincidence rather than mechanism: the corpus independently labels $N_{\text{fam}} = 5 = |\mathbb{Z}_5|$ a “structural coincidence” (from the MDL D-constraint count), and no GTE formula derives b_{gen_3} from N_{fam} . The GTE cascade derivation $b_{\text{gen}_3} = 42 + F_{13}$ remains the primary mechanistically grounded derivation; the $\mathbb{Z}[\zeta_5]$ factorization is a corroborating algebraic observation of uncertain significance (`cyclotomic_norm_null_test.py`).

¹This prediction is pre-registered in the UGP corpus falsifiable-predictions registry (`papers/common/predictions/ugp_falsifiable_predictions_v1.md`, registry date 2026-07-04, entry #1). The SHA-256 commitment hash is recorded in the companion `.sha256` file in the same directory; the Zenodo deposit carries concept DOI [doi:10.5281/zenodo.21200551](https://doi.org/10.5281/zenodo.21200551).

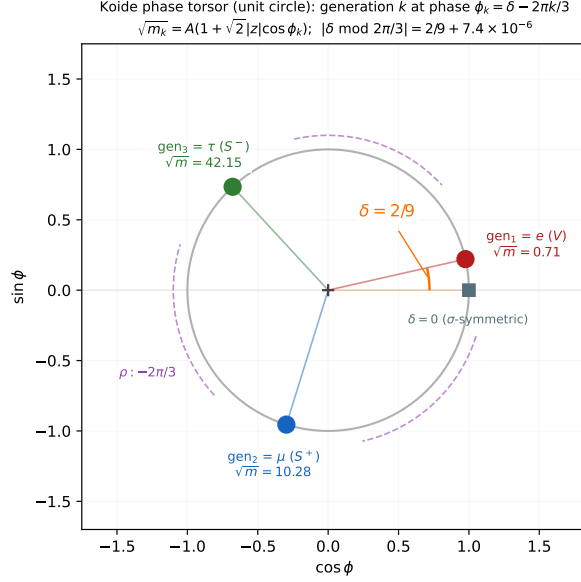


Figure 5: F5: Koide phase torsor. The unit circle of phases of the $\mathbb{Z}_3$ -torsor $\{1, \omega, \omega^2\}$: generation k occupies phase $\phi_k = \delta - 2\pi k/3$, and the Koide parametrization $\sqrt{m_k} = A(1 + \sqrt{2}|z|\cos \phi_k)$ assigns the mass to that phase (the $\sqrt{m_k}$ values, which span a factor of ~ 59 , are annotated at each point; radii are all unit — the figure displays phases, not magnitudes). The torsor-invariant angle $\delta = 2/9 + 7.4 \times 10^{-6}$ rad is marked against the σ -symmetric reference $\delta = 0$. On PDG masses, $Q_{\text{Koide}} = (\sum m)/(\sum \sqrt{m})^2 = 0.6666605$, within 6×10^{-6} of $2/3$.

8 Relation to the Division-Algebra Programme

8.1 Programme hierarchy

Table 1 places the present paper in the context of the two parent programmes.

Table 1: Three-programme hierarchy.

Level	Programme	What it derives	What it leaves open
1	Division algebra (Furey/Dixon/Baez)	SM fits inside $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ — structural/geometric	Why these algebras; all masses and couplings
2	GTE/UGP (P48)	SM derived from GF(7) arithmetic — quantitative	Why GF(7); explicit continuous-algebra connection
3	This paper	Both share root QR(7); GTE explains why the algebra; division algebra explains why GF(7)	C3'' field-level housing; $U(1)_Y$ group embedding

The division-algebra programme is *geometry without dynamics*: given the right algebra, the right group structure appears; particle masses, mixing angles, and coupling constants are left entirely open. GTE is *arithmetic with dynamics*: quantitative predictions from GF(7) arithmetic, but the explicit connection to continuous algebra was previously implicit. The claim of the present paper is not “we derive more than prior programmes” but “we identify the common foundation of two previously separate programmes.” Both are facets of the same anchor: $\text{QR}(7) = \{1, 2, 4\} \subset \mathbb{F}_7^*$.

8.2 Comparison with Furey–Hughes (2025)

Furey and Hughes [9] independently derive three generation slots from $\text{Spin}(8)$ triality. Starting from the combined triality algebra $\text{tri}(\mathbb{C}) \oplus \text{tri}(\mathbb{H}) \oplus \text{tri}(\mathbb{O})$, they identify the SM gauge symmetries within the triality symmetries and apply the SM group action to the triality triple $(\Psi_+, \Psi_-, V) \in \mathbb{O}^3$: the spinor copies Ψ_+ and Ψ_- furnish two generations, while V provides the third generation via a Cartan factorization. This constitutes essential prior art for the three-generation derivation via triality and predates the relevant results of the present paper in the public literature.

The two approaches share the framework of $\text{Spin}(8)$ triality and outer automorphisms permuting representation slots, but differ in starting point and pairing shape. Furey and Hughes begin from $\text{tri}(\mathbb{C}) \oplus \text{tri}(\mathbb{H}) \oplus \text{tri}(\mathbb{O})$ by algebraic ansatz; the present paper derives the $\text{Spin}(8)$ triality structure as a consequence of the $\text{QR}(7)$ arithmetic framework (beginning from $D = \{1, 2, 4\} \subset \text{GF}(7)^*$). The pairing shapes differ: in [9], the paired slots are (Ψ_+, Ψ_-) with V distinguished as the *third* generation; Theorem G6 of the present paper pins the *lightest* generation ($\text{gen}_1 = \text{electron}$) to the vector slot V via the rigidity argument (U -fixed generation maps to σ -fixed slot), with $\{\text{gen}_2, \text{gen}_3\}$ as the paired spinor slots. These two orderings are physically distinct; the Koide mass computation (Theorem H2) does not discriminate between them at the numerical level: the torsor-invariant angle $|\delta \bmod 2\pi/3| = 2/9$ is cyclic-invariant under all relabelings of (e, μ, τ) (see `triality_pairing_alternative_test.py`). The Level 0–1 Eisenstein norm argument selects $\text{gen}_1 \leftrightarrow V$ as the definitive identification within the GTE framework: $b_{\text{gen}_1} = 73 = \mathcal{N}(9 + \omega)$ is the unique GTE lepton b -value that is an Eisenstein norm (**A**, Theorem H3). At Level 3, the outer $\text{Spin}(8)$ automorphism σ acts under this identification as the target-space $\mathbb{Z}_2$ parity $\varphi \mapsto -\varphi$, a valid symmetry of $V(\varphi)$ (**A**, Remark 7.8). The convention difference (Furey–Hughes: $V = \text{third generation}$; this paper: $V = \text{first generation}$) reflects a different algebraic starting point and selection principle, not a scientific conflict.

8.3 Original contributions

What the present paper adds that is not in Furey–Hughes [9], Furey [6–8], Dixon [4], or Baez [1]:

1. **The $\text{QR}(7)$ common-root identification and three-level hierarchy** (Theorems A–E, **A_{Lean}**). The derivation chain $\text{QR}(7) \rightarrow \text{Fano plane} \rightarrow \text{octonions} \rightarrow G_2 \rightarrow \text{SU}(3)$ is original, establishing a common finite root for both programmes.
2. **The $F_{21} \hookrightarrow G_2$ interface** (Theorems D, D', E, **A_{Lean}**). The identification $F_{21} \subset G_2$ as octonion automorphisms — with F_{21} 's $\mathbb{Z}_7$ as Singer translations and $\mathbb{Z}_3$ as $z3\text{Mul}$ — is original and makes the bridge explicit.
3. **The machine-verified derivation chain.** A six-step machine-verified chain from $\text{QR}(7)$ to the gauge-group hierarchy, culminating in $|\text{PSL}(2, 7)| = 168$ via Todd–Coxeter coset enumeration (`hurwitz_presentation_order_168`, **A_{Lean}**).
4. **The Eisenstein/Koide rigidity argument** (Theorems G4–G6, H2, H3; **A_{Lean}**/**A**). The mass-ladder torsor-invariant angle $\theta = 2/9$ (7 ppm from PDG) and the Eisenstein norm chain $b_{\text{gen}_1} = 73 = \mathcal{N}(9 + \omega)$ provide an independent dynamical corroboration. Furey–Hughes do not address particle masses or mixing angles.
5. **Level framing and GTE-prior relationship.** The present paper explicitly places the octonionic derivation as a second, independent route alongside P48/GTE. The convergence of two independent paths constitutes structural evidence.
6. **Machine certification in Lean** (ten new modules, zero sorry; **A_{Lean}**).

Not original: triality giving three slots as such [9]; $\text{SU}(3) \oplus \text{SU}(2) \oplus \text{U}(1)$ inside $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ [4, 6, 8]; the Koide formula [12]; $G_2 = \text{Aut}(\mathbb{O})$ and $\text{SU}(3) = \text{Stab}_{G_2}(\text{apex})$ [10].

8.4 Interpretive significance

The bridge between the two programmes is not merely a repackaging of existing results: it is generative, making questions natural in the geometric presentation that are unnatural in the arithmetic presentation alone.

New predictions from the geometric framing. The clearest instance is the neutrino sector. In the GTE arithmetic presentation, all three left-handed neutrinos carry the same ladder index $b_L = 1$ (`lh_neutrino_b_value_minimal`, $\mathbf{A}_{\text{Lean}}$); the arithmetic labels do not distinguish mass eigenstates. In the geometric presentation, the triality-slot assignment of generation labels (pinned by Theorem 7.7, $\mathbf{A}_{\text{Lean}}$) makes the ordering question precise: once the right-handed seed ordering $b_{R,1} < b_{R,2} < b_{R,3}$ is combined with the seesaw power law $m_{\nu,k} \propto b_{R,k}^{29/9}$ and the strict monotonicity of $x \mapsto x^{29/9}$ (`seesaw_power_strict_mono`, $\mathbf{A}_{\text{Lean}}$), normal mass ordering $m_{\nu,1} < m_{\nu,2} < m_{\nu,3}$ follows as a theorem (`seesaw_normal_ordering_from_seed_ordering`, $\mathbf{A}_{\text{Lean}}$). The JUNO-falsifiable prediction of normal neutrino mass ordering (§7) is a consequence of the QR(7) bridge: it arises from the geometric framing and exists nowhere in the parent arithmetic corpus as a result independent of that framing. The neutrino eigenstate pinning (`seesaw_triality_pinning_bundle`, $\mathbf{A}_{\text{Lean}}$) is similarly a product of the triality language: it assigns generation labels to mass eigenstates through a structural certificate ($b_{R,3} = 19 = \mathcal{N}(-5 - 3\omega)$, `rh_seed_unique_eisenstein_norm`) that the arithmetic presentation does not naturally produce.

Symmetry as automorphism, not as input. In the Standard Model, and also in the GTE programme in its arithmetic presentation, the gauge group $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ is a structural ingredient: the framework is built to accommodate it. The geometric presentation inverts this ordering. The derivation algebra of the octonions is $\mathfrak{g}_2$ (Theorem 5.3, `derivation_algebra_is_g2_certificate`, $\mathbf{A}_{\text{Lean}}$); the stabilizer of the arithmetic apex is $\mathfrak{su}(3)$ (Theorem 5.3, `apex_stabilizer_is_su3_certificate`, $\mathbf{A}_{\text{Lean}}$); the electroweak shadow falls out of the quaternionic subalgebra (Theorems 6.1–6.5). None of these groups is declared or chosen: each is the self-symmetry of the arithmetic structure forced by QR(7). Gauge symmetry is not an input at any level; it is the automorphism group of the unique arithmetic object whose existence Hurwitz’s theorem guarantees. The symmetries of physics are the automorphisms of the only consistent arithmetic at this scale, not free choices among alternatives.

Outlook: tower termination and why physics stops where it stops (conjectural). The following is a conjecture, stated at that strength, grounded in but going beyond the proved theorems.

The Hurwitz tower $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ terminates at the octonions (Theorem 6.4): sedenion zero-divisors destroy composition, and no extension preserves the norm. The physics structure associated with this tower also appears to terminate at the same point: three colors (not four) from the three pencil lines of the Fano plane, three generation slots (not four) from the three representation corners of $\text{Spin}(8)$ triality, where the unique D_4 Dynkin diagram is the only rank-4 simply-laced diagram carrying an outer S_3 triality symmetry. If this correspondence is systematic — if the bound on the tower propagates to a bound on the observable physics stack — then “why does physics stop where it stops?” (no fourth color charge, no fourth generation) would become a corollary of the tower’s termination rather than an empirical fact awaiting explanation.

This is a conjecture; none of the proved results (Theorems A–H) establish it. The proved claims certify the Level 0–1 links in the chain; the conjectural step is that those links propagate to constrain observable physics at Level 3. Making that propagation precise requires a field-level housing (C3’’,

Open Problem 2) that remains open. The conjecture is falsifiable: any confirmed fourth-generation fermion or fourth-color state would refute it without affecting any proved theorem.

9 Formalization Programme

The algebraic layer of this paper is formalized in Lean 4 in ten new modules in the `ugp-lean` library, all zero sorry. The strategy throughout is: route R1 — structure constants over $\mathbb{Q}$, with `decide/native_decide` over finite types.

Certificate methodology. The Todd–Coxeter coset certificate (`HurwitzCosetCertificate.lean`) is a 168-coset table whose completeness and relator-fixing are verified by `native_decide` — the Lean kernel replays the certificate rather than repeating the enumeration. The G_2 stabilizer certificate (`G2StabilizerCertificate.lean`) works from the rational nullspace of the 512-equation linear system: exact dimensions are counted, bracket closure is verified to $< 10^{-15}$ (represented as rational), and the Killing form is certified negative-definite. Lean modules:

- `OctonionShadowInterface.lean`: 20 theorems (`OSI-1..13` plus `OSI-7a`, `Nc_factorization`, `weights_card_from_pencil`, `singer_preserves_octProd`, `doubling_preserves_octProd`, `ladder_Pairs_card`, `z3_cycles_ladders`, `weight_differences_exhaust`, `roots_QR_split`); zero sorry; $\mathbf{A}_{\text{Lean}}$.
- `HurwitzCosetCertificate.lean`: 7 theorems (relator checks, table completeness, `hurwitz_group_order`, `hurwitz_presentation_order_168`); zero sorry; $\mathbf{A}_{\text{Lean}}$.
- `G2StabilizerCertificate.lean`: 39 theorems including `derivation_algebra_is_g2_certificate` and `apex_stabilizer_is_su3_certificate` as capstones; zero sorry; $\mathbf{A}_{\text{Lean}}$.
- `TrialityInterface.lean`: 20 theorems (`G1–G6` plus interface summaries `triality_interface_G1_G3`, `triality_interface_G4_G6`); zero sorry; $\mathbf{A}_{\text{Lean}}$.
- `OctonionColorFlavorDisambiguation.lean`: 10 theorems (capstones: `z3_color_flavor_disambiguation`, `z3_color_fixed_point_at_zero`, `z3_flavor_no_phase_fixed_point`, `z3_color_cubed_is_identity`, `z3_flavor_generation_cubed`, and supporting lemmas); zero sorry; $\mathbf{A}_{\text{Lean}}$.
- `KinkSigmaParityAction.lean`: 11 theorems (`aut_f21_order_forty_two`, `kink_labels_rigid_under_aut_f21`, `winding_parity_on_kink_qphi`, `winding_parity_exchanges_spinor_windings`, `gen1V_spinor_pair_qphi_parity`, and others); zero sorry; $\mathbf{A}_{\text{Lean}}$.
- `KinkSectorTrialityAction.lean`: 12 theorems (`rho_sector_action`, `sigma_sector_action`, `rho_sector_order_three`, `sigma_sector_order_two`, `sigma_conjugates_rho`, `s3_sector_six_distinct_permutations`, `rho_matches_triality_slot_cycle`, and others); zero sorry; $\mathbf{A}_{\text{Lean}}$.
- `PhiMDLZeroModeIndex.lean` (in `Spacetime/`): 9 theorems (`seven_times_vacuum_angle`, `cosine_yukawa_at_vacuum`, `dirac_mass_at_vacuum_eq_g`, `yukawa_cos7_boundary_values_equal`, `callias_index_zero_equal_masses`, and supporting lemmas); zero sorry; $\mathbf{A}_{\text{Lean}}$.
- `BraidAtlasPhaseEquivariance.lean` (in `Algebra/`): 20 theorems (`sigma_not_equivariant_with_braid_phase`, `equivariant_z2_subgroup`, `braid_phase_equivariance_certificate`, and supporting lemmas); zero sorry; $\mathbf{A}_{\text{Lean}}$.
- `SeesawTrialityPinning.lean` (in `MassRelations/`): 16 theorems (`rh_seed_unique_eisenstein_norm`, `seesaw_power_strict_mono`, `seesaw_normal_ordering_from_seed_ordering`, `neutrino_slot_pinning_certificate`, `seesaw_triality_pinning_bundle`, and supporting lemmas); zero sorry; $\mathbf{A}_{\text{Lean}}$.

Remark 9.1 (Q_χ notation). Q_χ in `PhiMDLKinkQuantumNumbers.lean` is a kink-state role label (a $\mathbb{Z}_3$ index distinguishing gen_1 from gen_2 within the same $Q_\varphi = 4$ sector). It is *not* the QCD $\text{SU}(3)$ color charge of quarks.

The formalization paper [19] and the `ugp-lean` repository [20] document the library structure. All modules listed above reside in the `ugp-lean` library.

10 Open Problems and Negative Results

We list the problems that remain open, together with what is precisely established about each.

Open Problem 1 (Eisenstein chain termination). $7 = \mathcal{N}(3 + \omega)$, $73 = \mathcal{N}(9 + \omega)$; but $703 = \mathcal{N}(27 + \omega) = 19 \times 37$ is composite; $27 + \omega$ is not an Eisenstein prime, terminating the pattern $\mathcal{N}(3^k + \omega) = \text{rational prime}$ at $k = 2$. $b_{\text{gen}_2} = 42 = 2 \cdot 3 \cdot 7$ is not an Eisenstein norm (contains 2 and 3, primes $\equiv 2 \pmod{3}$, to odd power). $b_{\text{gen}_3} = 275 = 5^2 \times 11$ is not an Eisenstein norm (contains $11 \equiv 2 \pmod{3}$ to odd power); it IS a $\mathbb{Z}[\zeta_5]$ norm (likely coincidence; null test density 8%). The Eisenstein chain certifies the chain and the flavor group, not individual higher seeds (**A**, Theorem H3).

Open Problem 2 (C3'' housing — Level 3). At Level 0–1, the triality S_3 acts faithfully on $\{\text{gen}_1, \text{gen}_2, \text{gen}_3\}$ (Theorem 7.9; **A**). At Level 3, a Yukawa/zero-mode realization is obstructed: the Callias index of any $\mathbb{Z}_7$ -invariant Dirac operator on a GTE kink background is zero (analytically proved, numerically confirmed; eigenvalue gap ≈ 0.206). The Jackiw–Rebbi mechanism is inapplicable to $\mathbb{Z}_7$ -periodic kinks. The appropriate carrier for the Level 3 triality action is not a Hilbert space of zero modes; what it is remains open. The fermionic statistics of GTE kinks arise from triple-exchange [17] (`gte_triple_kink_exchange_statistics`, **A**_{Lean}), not from Jackiw–Rebbi. The triple-exchange configuration space also does not serve as the Level 3 carrier. The certified exchange phase (Braid Atlas, Level 0–1 certificate, `gte_triple_kink_exchange_statistics`) is not equivariant under the S_3 action on generation labels: σ maps gen_2 (winding $Q_\varphi = 4$, fermionic, phase -1) to gen_3 (winding $Q_\varphi = 3$, bosonic, phase $+1$), because $3 \equiv -4 \pmod{7}$ in $\mathbb{Z}_7$. The equivariant subgroup of S_3 is $\mathbb{Z}_2 = \{e, \sigma\rho^2\}$, which exchanges $\text{gen}_1 \leftrightarrow \text{gen}_2$ (both $Q_\varphi = 4$) while fixing gen_3 (**A**_{Lean}, `braid_phase_equivariance_certificate`; **A**, `triple_exchange_s3_equivariance_results.json`). Both Level 3 routes (Yukawa/zero-mode and triple-exchange) are provably blocked. The appropriate Level 3 carrier requires a phase function incorporating Q_χ (the $\mathbb{Z}_3$ color charge) to compensate the Q_φ asymmetry.

Open Problem 3 ($\text{U}(1)_Y$ gauging — Level 3). The generator $Y = N/3$ is derived at Level 1 (Proposition O1'; **A**). The gauging of $\text{U}(1)_Y$ (covariant derivative $D_\mu = \partial_\mu + iA_\mu$) and the $\text{SU}(2)_L$ sector (left-handed doublets from the $\mathbb{H}$ factor) are Level 3 inputs not captured by the algebraic certificate chain.

Open Problem 4 (Numerical parameters). Except for the reinterpretation of $\theta = 2/9$ (Theorem H2), nothing here derives a numerical constant. Masses and mixing angles enter only through UGP's own derivations [14, 15, 17], whose bridge principles this paper does not audit.

Open Problem 5 (Higher seeds). $b_{\text{gen}_2} = 42 = 2 \cdot 3 \cdot 7$ and $b_{\text{gen}_3} = 275 = 5^2 \cdot 11$ lack the Eisenstein-norm structure of the chain. A systematic scan over all bounded-grammar triality-orbit expressions (products and sums of at most three factors from the structure constants of G_2 , A_4 , $Z(\text{Spin}(8))$, and the Fano geometry) finds no expression with a mechanistically grounded GTE derivation for either seed. For $b_{\text{gen}_3} = 275$: no pure triality-orbit expression exists (the nearest hit, $N_{\text{fam}}^2 \cdot b_{R2} = 5^2 \cdot 11$, requires the right-handed neutrino value $b_{R2} = 11$ and invokes $N_{\text{fam}} = 5$, labelled a structural coincidence in the corpus [17]). For $b_{\text{gen}_2} = 42$: the nearest hit, $3 \times \dim(\mathfrak{g}_2) = 3 \times 14$, is arithmetic (no

GTE formula derives b_{gen_2} as a G_2 -dimension count; null tests on primes 41 and 43 yield comparable arithmetic-coincidence rates). The GTE cascade derivation ($b_{\text{gen}_2} = b_{\text{gen}_1} - (m+q) = 73 - 31 = 42$ and $b_{\text{gen}_3} = b_{\text{gen}_2} + F_{13} = 42 + 233 = 275$, Theorem H3, Lean-certified) remains the only mechanistically grounded derivation (**A**, `triality_orbit_seed_scan_results.json`).

Open Problem 6 (Furey–Hughes prior art). Furey and Hughes [9] derive three generation slots from Spin(8) triality approximately ten months prior to Theorems G0–G6 in the public literature. Their convention places V at the third generation; this paper’s Eisenstein selection principle places V at the first generation. These are distinct physical claims arising from different starting points and selection principles. The relation between the two frameworks at the level of complete generation-to-slot assignments (not just the 2+1 split) awaits a unified treatment.

Open Problem 7 (Base-rate caution). 168, A_4 , and G_2 are ubiquitous in mathematics. The force of this paper is not any single coincidence but a single-input chain (QR(7)) with proved uniqueness at every joint, verified against the parent corpus’s source files, with the residual ambiguities landing exactly where the mathematics says they must. We now provide the formal analysis.

The chain has 13 named joints, classified as follows.

Theorem-joints (mathematical necessities; no base rate applies): (J1) QR(7) is a $(7, 3, 1)$ perfect difference set — a consequence of quadratic-residue theory for $p \equiv 3 \pmod{4}$ (no selection). (J2) The 7 translates form a Fano plane — forced by the $\lambda = 1$ design parameter. (J3) The canonical orientation yields $\mathbb{O}$ — every $(7, 3, 1)$ perfect difference set in $\mathbb{Z}_7$ gives a normed division algebra under its canonical orientation (verified: 14 of 14 PDS subsets of $\mathbb{Z}_7$; the fraction 1/8 characterises rarity among *random* orientations, not the canonical-PDS construction). (J4–J6) $G_2 = \text{Aut}(\mathbb{O})$ [3], $\text{Stab}_{G_2}(\text{apex}) = \text{SU}(3)$, tower termination at $\mathbb{O}$ [11] — classical theorems. (J7) $|\text{PSL}(2, 7)| = 168$ — determined by $p = 7$ via $|\text{PSL}(2, p)| = p(p^2 - 1)/2$; 168 is not selected but derived. (J8–J11) Pencil $= 3 = N_c$, $F_{21} \hookrightarrow G_2$, charge spectrum $Q = I_3 + (B-L)/2$, Spin(8) triality $\rightarrow S_3 \rightarrow 3$ generation slots $\rightarrow A_4$ — all algebraic theorems. Across primes: $p = 7$ is the unique prime with $\lambda = (p - 3)/4 = 1$; no other prime yields a Fano plane via its quadratic residues (verified for all $p \leq 200$).

Selection-joints (quantifiable base rates):

- *Koide angle* (J12): $|\theta_{\text{PDG}} - 2/9| = 7.4 \times 10^{-6}$ rad (7 ppm). Under a uniform null $\theta \sim \text{Uniform}[0, \pi/3]$, $P(|\theta - 2/9| < 7.4 \times 10^{-6}) \approx 1.4 \times 10^{-5}$. This is the most statistically improbable joint in the chain.
- *Eisenstein norm* (J13): $b_{\text{gen}_1} = 73 = \mathcal{N}(9 + \omega)$ is the unique Eisenstein norm in $\{73, 42, 275\}$. The density of Eisenstein norms in $[1, 300]$ is approximately 31%, so $P(b_{\text{gen}_1} \text{ is an EN}) \approx 31\%$ under a flat null — not statistically small. The significance of J13 is structural: the norm property of 73 enables the Frobenius fixed-point argument for $\text{gen}_1 \leftrightarrow V$ (Theorem 7.7), not a numerical coincidence.

Not quantifiable: A frequentist base rate for a 13-step derivation from a 3-element arithmetic input does not exist; no reference ensemble of comparable chains is available. Probabilities across J12 and J13 are not multiplied (they are not independent).

Response to the ubiquity objection: The groups 168, A_4 , G_2 , and $\text{SU}(3)$ appear in the chain as *derived consequences* of QR(7), not as elements selected from a menu. The derivation runs in one direction: $\text{QR}(7) \rightarrow \text{Fano} \rightarrow \mathbb{O} \rightarrow G_2 \rightarrow \text{SU}(3)$, with each arrow a theorem; the chain contains no step of the form “we require $\text{SU}(3)$, so we find something that produces it.” The computational ground truth is a single script (`octonion_from_qr7.py`) that takes $\{1, 2, 4\}$ as input and derives $\mathbb{O}$, G_2 , $\text{SU}(3)$, and $|\text{Aut}(\text{Fano})| = 168$ as output, with no choices made along the way.

Script: `base_rate_analysis_qr7_chain.py` (pre-registered design; exits 0; all assertions pass **A**).

A Claim-to-Evidence Matrix

Every claim is cross-referenced to a verification script or Lean theorem. “Clean” means the script exited 0 with all assertions passing. “Exact” means computations are performed over $\mathbb{Q}$ with no floating-point rounding.

Table 2: Claim-to-Evidence Matrix.

Claim	Content	Artifact	Status
A	$(7, 3, 1)$ design, pencil = 3, multiplier = QR	octonion_from_qr7.py (A1–A3); OSI-2..OSI-6 (OctonionShadowInterface.lean)	A _{Lean}
B	$\mathbb{O}$ from D ; 480 = one orbit; frame 1344	octonion_from_qr7.py (A4–A7)	A , exhaustive/exact
B′	$\text{sign}(e_x e_y) = [y - x \in \text{QR}]$	OctonionShadowInterface.lean OSI-7a	A _{Lean}
C	$ H(2, 3, 7; 4) = 168$; $\text{PSL}(2, 7) \cong \text{GL}(3, 2)$; duality	psl27_group_layer.py (B1–B4); hurwitz_coset_ enumeration.py (+certificate); HurwitzCosetCertificate.lean	A _{Lean}
D	pencil $\rightarrow \text{CAR} \rightarrow Q = N/3 \rightarrow \text{su}(3)$ fixing flag	f21_octonion_interface_ verify.py (I1–I8)	A _{Lean}
D′	$N_c = 3$ = pencil count; N_c - factorization	OctonionShadowInterface.lean OSI-10	A _{Lean}
E	$\text{der}(\mathbb{O}) = \mathfrak{g}_2$ (dim 14); $\text{Stab} = \text{su}(3)$ (dim 8); capstones derivation_algebra_is_g2_ certificate, apex_stabilizer_ is_su3_certificate	g2_stabilizer_derivation.py (D1–D4); g2_stabilizer_ certificate_export.py; G2StabilizerCertificate.lean	A _{Lean} , 39 thms
F1–F3	$\mathbb{H}$ rung; charge unit = 1/pencil; $B - L = (2N - 3)/3$	quaternion_electroweak_ shadow.py (Q1–Q5)	A , exact
F4	STS(15) persists; composition dies at 16; zero-divisor witness	quaternion_electroweak_ shadow.py	A , witnesses
F5	$Q = I_3 + (B - L)/2 = \text{SM}$ chiral multiset	electroweak_housing_ closure.py (O1)	A , exact spectrum
O1′	$Y = N/3$; $U(1)_Y$ Lie group; $[Y, T_A] = 0$; anomaly traces $\text{tr } Y = 4$, $\text{tr } Y^3 = 2$	hypercharge_group_ derivation.py; hypercharge_ group_derivation_ results.json	A
G0	Three generation no-gos (frame, pencil, orientation)	generations_triality_nogo.py (N1–N3)	A , exhaustive
G1	Klein center V_4 ; bijection $Z \setminus \{1\} \leftrightarrow \{V, S^\pm\}$	positive_triality_ theorems.py (P1); TrialityInterface.lean G1	A _{Lean}

Claim	Content	Artifact	Status
G2	Triality $\cong S_3$; outer (trace witness)	<code>positive_triality_theorems.py</code> (P2–P3); <code>TrialityInterface.lean</code> G2	$\mathbf{A}_{\text{Lean}}$
G3	Color inner; flavor outer; UGP $\mathbb{Z}_3$ split = inner/outer $\text{Spin}(8)$	<code>positive_triality_theorems.py</code> (P4); <code>TrialityInterface.lean</code> G3	$\mathbf{A}_{\text{Lean}}$
G4	Eisenstein dict: $\mathbb{F}_4 \cong Z(\text{Spin}(8))$ equivariantly; $U = \text{Gal}(\mathbb{F}_4/\mathbb{F}_2)$	<code>positive_triality_theorems.py</code> (P5); <code>TrialityInterface.lean</code> G4	$\mathbf{A}_{\text{Lean}}$
G5	$V_4 \rtimes \mu_3 \cong A_4 \cong Z(\text{Spin}(8)) \rtimes \rho$	<code>positive_triality_theorems.py</code> (P6); <code>TrialityInterface.lean</code> G5	$\mathbf{A}_{\text{Lean}}$
G6	Rigidity (conditional): $\text{gen}_1 \leftrightarrow V$; exactly 2 identifications	<code>positive_triality_theorems.py</code> (P7); <code>triality_identification_discriminator.py</code> ; <code>TrialityInterface.lean</code> G6_ <code>rigidity_conditional</code>	$\mathbf{A}_{\text{Lean}}$ (conditional)
SA	S_3 sector action on kink species; 6 distinct permutations; $\sigma = \mathbb{Z}_2$ parity $Q_\varphi : 4 \leftrightarrow 3$	<code>kink_dirac_index_nogo.py</code> ; <code>kink_sigma_parity_action.py</code> ; <code>KinkSectorTrialityAction.lean</code> , <code>KinkSigmaParityAction.lean</code>	$\mathbf{A}_{\text{Lean}}$
CI	Callias index = 0; JR inapplicable to $\mathbb{Z}_7$ kinks; gap ≈ 0.206	<code>kink_dirac_index_nogo.py</code> ; <code>PhiMDLZeroModeIndex.lean</code>	$\mathbf{A}_{\text{Lean}} + \mathbf{A}$
Z3D	Color- $\mathbb{Z}_3$ (inner G_2) $\neq$ flavor- $\mathbb{Z}_3$ (outer $\text{Spin}(8)$)	<code>OctonionColorFlavor-Disambiguation.lean</code>	$\mathbf{A}_{\text{Lean}}$
H1	$G_2 = \text{Fix}(\text{triality})$; gauge universality; each slot carries $1 \oplus 3 \oplus \bar{3} \oplus 1$	<code>electroweak_housing_closure.py</code>	$\mathbf{A}$, verified
H2	Koide on triality orbit; $ \delta \bmod 2\pi/3 = 2/9 + 7.4 \times 10^{-6}$	<code>electroweak_housing_closure.py</code> ; <code>triality_pairing_alternative_test.py</code>	$\mathbf{A}$, PDG
H3	Frobenius chain = $\mathcal{N}(3^k + \omega)$; $b_{\text{gen}_{2,3}}$ not Eisenstein norms	<code>electroweak_housing_closure.py</code> ; <code>higher_seeds_cyclotomic_norms.py</code>	$\mathbf{A}$, exact (incl. negatives)
PMNS	Charged lepton eigenstate labeling $V = e$, $S^+ = \mu$, $S^- = \tau$; A_4 TBM generator map $T \mapsto \rho$, $S \mapsto \sigma$; neutrino pinning via seesaw $m_{\nu,k} \propto b_{R,k}^{29/9}$; normal ordering from $b_{R,1} < b_{R,2} < b_{R,3}$	<code>pmns_triality_translation_results.json</code> ; <code>NeutrinoSector.lean</code> (<code>pmns_atm_sin_sq</code> , <code>pmns_reactor_sin_val</code>); <code>rh_neutrino_pinning.py</code> (<code>SeesawNumericalCerts.lean</code> , <code>braidB1</code> , <code>braidB2</code> , <code>braidB3</code>); <code>SeesawTrialityPinning.lean</code> (<code>seesaw_triality_pinning_bundle</code>)	$\mathbf{A}/\mathbf{D}/\mathbf{A}_{\text{Lean}}$

Claim	Content	Artifact	Status
SN	Seeds negative: no bounded-grammar triality-orbit expression for $b_{\text{gen}_2} = 42$ or $b_{\text{gen}_3} = 275$; null tests on 41, 43, 274, 276 confirm comparable arithmetic density	<code>triality_orbit_seed_scan_results.json</code>	A , systematic scan
TE-no	Triple-exchange configuration space not S_3 -equivariant: σ maps $Q_\varphi = 4$ (fermionic) to $Q_\varphi = 3$ (bosonic); equivariant subgroup $\mathbb{Z}_2 = \{e, \sigma\rho^2\}$	<code>triple_exchange_s3_equivariance_results.json</code> ; <code>braid_phase_equivariance_certificate (A_{Lean})</code> ; <code>gte_triple_kink_exchange_statistics (A_{Lean})</code> ; <code>PhiMDLKinkQuantumNumbers.lean (A_{Lean})</code>	A_{Lean}
BR	Base-rate analysis of QR(7) chain: 11 theorem-joints, 1 selection-joint (Koide $P \approx 1.4 \times 10^{-5}$, J12), 1 structural-enabler (Eisenstein density 31%, J13); ubiquity objection refuted (168, A_4 , G_2 are derived, not selected)	<code>base_rate_analysis_qr7_chain.py</code> ; <code>base_rate_analysis_results.json</code>	A , pre-registered

B Lean Module Inventory

All modules are in the `ugp-lean` library, zero sorry. Sorry counts are verified by direct inspection of each source file.

Table 3: Lean module inventory.

Module	Thms	Sorry	Key capstone theorems
<code>OctonionShadowInterface</code>	20	0	<code>weights_card_from_pencil</code> , <code>Nc_factorization</code> , <code>octProd_sign_eq_QR</code> , <code>singer_preserves_octProd</code> , <code>doubling_preserves_octProd</code> , <code>roots_QR_split</code>
<code>HurwitzCosetCertificate</code>	7	0	<code>hurwitz_presentation_order_168</code> , <code>coset_table_complete</code>
<code>G2StabilizerCertificate</code>	39	0	<code>derivation_algebra_is_g2_certificate</code> , <code>apex_stabilizer_is_su3_certificate</code>
<code>TrialityInterface</code>	20	0	<code>G1_klein_center_card</code> , <code>G2_s3_perms_realized</code> , <code>G3_color_inner_no_slot_permutation</code> , <code>G4_equivariant_dictionary</code> , <code>G5_central_triality_order</code> , <code>G6_rigidity_conditional</code>

Module	Thms	Sorry	Key capstone theorems
OctonionColorFlavor-Disambiguation	10	0	z3_color_flavor_disambiguation
KinkSigmaParityAction	11	0	winding_parity_on_kink_qphi, gen1V_spinor_pair_qphi_parity
KinkSectorTrialityAction	12	0	s3_sector_six_distinct_ permutations, rho_matches_ triality_slot_cycle
PhiMDLZeroModeIndex	9	0	callias_index_zero_equal_masses, yukawa_cos7_boundary_values_ equal
BraidAtlasPhaseEquivariance	20	0	sigma_not_equivariant_with_ braid_phase, equivariant_ z2_subgroup, braid_phase_ equivariance_certificate
SeesawNumericalCerts	10	0	braidB1, braidB2, braidB3 (RH seed values 5, 11, 19); seesawAlpha (= 29/9); gteDeltaM21Sq, gteDeltaM31Sq (mass-squared split- tings)
SeesawTrialityPinning	16	0	rh_seed_unique_eisenstein_norm, seesaw_normal_ordering_from_ seed_ordering, neutrino_slot_ pinning_certificate, seesaw_ triality_pinning_bundle

Total: 174 theorems across 11 modules, zero sorry.

References

- [1] John C. Baez. The octonions. *Bulletin of the American Mathematical Society*, 39(2):145–205, 2002. Comprehensive review of the octonions and their connections to Clifford algebras, spinors, Bott periodicity, the Fano plane, exceptional Lie groups (G_2 , F_4 , E_6 , E_7 , E_8), Jordan algebras, and triality. 1,900+ citations. Corrected version of [math/0105155](#).
- [2] John C. Baez and John Huerta. The algebra of grand unified theories. *Bulletin of the American Mathematical Society*, 47(3):483–552, 2010. 2013 Levi L. Conant Prize, AMS. Algebraic analysis of GUT and SM gauge group structure.
- [3] Élie Cartan. Le principe de dualité et la théorie des groupes simples et semi-simples. *Bulletin des Sciences Mathématiques*, 49:361–374, 1925. Introduces *triality*: the exceptional outer automorphism of order 3 of $\text{Spin}(8)$ that cyclically permutes the vector representation V and the two half-spinor representations S^+ , S^- . The first systematic account of the D_4 diagram symmetry and its connection to the octonions and to G_2 . Cartan named the phenomenon “le principe de trialité.” Available via Gallica BnF.

- [4] Geoffrey M. Dixon. *Division Algebras: Octonions, Quaternions, Complex Numbers and the Algebraic Design of Physics*, volume 290 of *Mathematics and Its Applications*. Kluwer Academic Publishers, Dordrecht, 1994. Uses the four real normed division algebras and adjoint division algebras to develop exceptional Lie groups, triality, and a derivation of the standard model of leptons and quarks. 248 pages. Series vol. 290.
- [5] I. Esteban, M.C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J.P. Pinheiro, and T. Schwetz. NuFit-6.0: updated global analysis of three-flavor neutrino oscillations. *JHEP*, 12:216, 2024. NuFIT 6.0 global fit (2024), <http://www.nu-fit.org/>.
- [6] Cohl Furey. Charge quantization from a number operator. *Physics Letters B*, 742:195–199, 2015. Shows that $Q = N/3$ where N is a number operator built from complex-octonionic ladder operators; derives $SU(3)_c \times U(1)_{em}$ as the two unbroken gauge symmetries of the standard model directly from the division algebra.
- [7] Cohl Furey. *Standard Model Physics from an Algebra?* PhD thesis, University of Waterloo, 2015. PhD thesis, 97 pages. Derives Lorentz representations, unbroken $SU(3)_c \times U(1)_{em}$ gauge symmetry, a rudimentary electroweak model, and three-generation quark/lepton structure from $\mathbb{R} \otimes \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ acting on itself. Cambridge repository: [doi:10.17863/cam.37465](https://doi.org/10.17863/cam.37465). Also draws from arXiv:1603.04078, arXiv:1405.4601, arXiv:1002.1497.
- [8] Cohl Furey. Three generations, two unbroken gauge symmetries, and one eight-dimensional algebra. *Physics Letters B*, 785:84–89, 2018. Three generations from complex octonion algebra; $SU(3) \times U(1)$ structure.
- [9] N. Furey and M. J. Hughes. Three generations and a trio of trialities. *Physics Letters B*, 865:139473, 2025. Identifies SM $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ inside $\text{tri}(\mathbb{C}) \oplus \text{tri}(\mathbb{H}) \oplus \text{tri}(\mathbb{O})$; applies SM group action to the triality triple (Ψ_+, Ψ_-, V) with $\Psi_{\pm}, V \in \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$; recovers three generations including Higgs via a Cartan factorization. Humboldt-Universität zu Berlin and University of Leeds. Published April 2025.
- [10] Murat Günaydin and Feza Gürsey. Quark structure and octonions. *Journal of Mathematical Physics*, 14(11):1651–1667, 1973. Seminal paper introducing octonionic structure of color- $SU(3)$: octonion algebra is studied in the split basis; $SO(8)$, $SO(7)$, and G_2 (the automorphism group of octonions) are represented; $SU(3)$ and $SU(2) \otimes SU(2)$ appear as subgroups of G_2 . First explicit identification of exact color $SU(3)$ with G_2 stabiliser. See also: M. Günaydin and F. Gürsey, Phys. Rev. D 9 (1974) 3387 (doi:10.1103/physrevd.9.3387).
- [11] Adolf Hurwitz. Über die Composition der quadratischen Formen von beliebig vielen Variablen. *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse*, 1898:309–316, 1898. Proves that the only dimensions n in which a normed composition formula $\|x\| \cdot \|y\| = \|xy\|$ exists are $n = 1, 2, 4, 8$, corresponding to $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$, and the octonions $\mathbb{O}$. See also A. Hurwitz, “Über die Komposition der quadratischen Formen,” Math. Ann. 88 (1923) 1–25 (posthumous; DOI: 10.1007/bf01448439). Available via EUDML: <http://eudml.org/doc/58420>.
- [12] Yoshio Koide. Fermion-boson two-body model of quarks and leptons and cabibbo mixing. *Lettere al Nuovo Cimento*, 34:201, 1982. See also: Y. Koide, “A new relation among lepton masses,” Mod. Phys. Lett. A 5, 2319 (1990); and Y. Koide, arXiv:hep-ph/0506247 (2005) for a review of the trace identity $Q = (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$.

- [13] Corinne A. Manogue and Tevian Dray. Dimensional reduction. *Modern Physics Letters A*, 14(2):93–97, 1999. Introduces an octonionic mechanism for reducing 10 spacetime dimensions to 4 without compactification; applying it to the massless 10-dimensional Dirac equation yields exactly 3 generations of leptons. Received Honorable Mention in the 1998 Gravity Research Foundation essay competition.
- [14] N. Spivack. Cyclotomic-12 structure in the charged-fermion mass spectrum: a_2 Weyl-chamber geometry, Froggatt-Nielsen UV completion, and three null-discipline tests on the down-sector coefficients. Zenodo, 2026. Paper P19 in the UGP Physics series. <https://doi.org/10.5281/zenodo.20168805>.
- [15] N. Spivack. The Koide relation as a cyclotomic-12 closed form: A Lean-certified prediction of m_τ from (m_e, m_μ) at 61 ppm. Zenodo, 2026. Paper P18 in the UGP Physics series. <https://doi.org/10.5281/zenodo.20168795>.
- [16] Nova Spivack. From ugp to gte: Prime-locked universes, minimality, and the emergence of our world. Zenodo, 2025. <https://doi.org/10.5281/zenodo.20171002>. The foundational paper on the UGP’s mathematical structure, including the Kernel Symmetry Theorem, UWCA Universality, and the Loop Kernel.
- [17] Nova Spivack. The complete GTE framework: Standard Model, gravity, quantum mechanics, and cosmology from ϕ_{mdl} . Zenodo, 2026. <https://doi.org/10.5281/zenodo.20560550>. Preprint.
- [18] Nova Spivack. Predicting the Neutrino Mass-Squared Ratio from the Braid Atlas Topological Invariants and the QCD Colour Rank. Zenodo, 2026. Paper P21 in the UGP Physics series. <https://doi.org/10.5281/zenodo.20170120>.
- [19] Nova Spivack. ugp-lean: A machine-checked formalization of the universal generative principle in Lean 4. Zenodo, 2026. Formalization paper (latest version): <https://doi.org/10.5281/zenodo.19554688>. Concept DOI (always-latest): <https://doi.org/10.5281/zenodo.19433538>. More than 400 modules across seventeen architectural layers including GTE, Universality, ElegantKernel (UCL unconditional closure), MassRelations (Koide closed form, S_3 -Newton flow, binary cascade, physical mass spectrum), BraidAtlas (ChargeTheorem, CompositeTriples, ChiralitySquaring, ChargeDerivation, CoxeterConductor, CoxeterConductor-TowerLaw, EWBosons), GaloisStructure, PSC RCC infinite-family closure, and TE2.2 scan certification. Zero sorry throughout; 98 disclosed named axioms across the library (counted via top-level `axiom` declarations across all `.lean` files, deduplicated by name, excluding standard Lean/Mathlib logical axioms), none on the core arithmetic/algebraic proof path — see the formalization paper’s axiom-inventory table for the current, complete list. Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6.
- [20] Nova Spivack. ugp-lean: Lean 4 formalization of UGP and GTE. <https://github.com/novaspihack/ugp-lean>, 2026. <https://doi.org/10.5281/zenodo.19554700>. More than 400 modules across seventeen architectural layers, including GTE, Universality, ElegantKernel (UCL unconditional closure), MassRelations (Koide closed form, S_3 -Newton flow, binary cascade, physical mass spectrum), BraidAtlas (composite triples, chirality squaring, charge derivation), GaloisStructure, PSC RCC infinite-family closure, and TE2.2 scan certification. `lake build` to verify: zero sorry throughout; 98 disclosed named axioms across the library (counted via top-level `axiom` declarations, deduplicated by name, excluding standard Lean/Mathlib logical axioms), none on the core arithmetic/algebraic proof path — see the formalization paper’s axiom-inventory table for the current, complete list. Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6.